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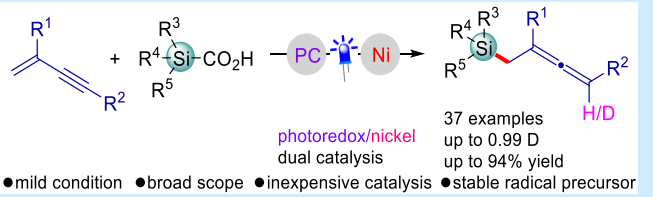
Regioselective Hydro(deutero)silylation of 1,3-Enynes Enabled by Photoredox/Nickel Dual Catalysis

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ABSTRACT: In the presence of visible light irradiation, organo-photo/nickel dual catalysts, and the mild base K_2HPO_4, 1,3-enynes react with silanecarboxylic acids to give the corresponding α-silylallenes with high selectivity. In this uniquely decarboxylative hydrosilylation of 1,3-enynes, a silyl radical process is involved and diverse electron-rich and -poor substrates proceed smoothly in moderate to excellent yields. This transformation is particularly synthetically worthwhile when using MeOD as the solvent, which furnishes new access to α-silyldeuteroallenes.



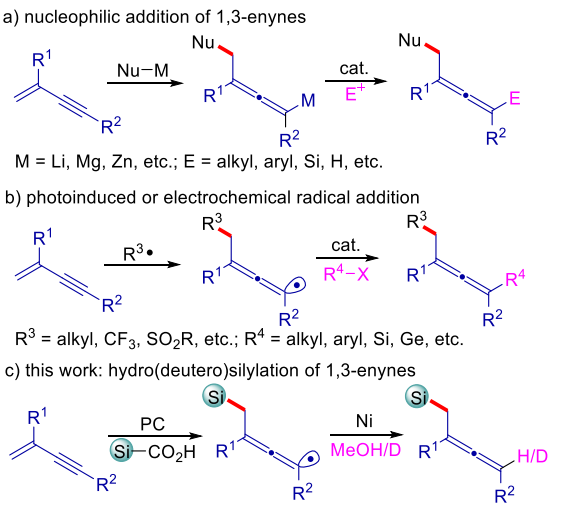
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Allene units not only widely exist in natural products, pharmaceuticals, bioactive molecules, and functional materials[1] but are also valuable intermediates in many organic transformations.[2] The first example of allene synthesis could be traced back to 1887 by Burton and Pechmann;[3] since then, methods for the preparation of allenes have been well-developed,[4] such as tropic rearrangement of propargylics,[5] prototropic isomerization of alkynes,[6] and elimination of enolphosphates[7] or vinylsulfoxide.[8] However, limitations exist with such approaches, which always rely on harsh reaction conditions or tedious prefunctionalization of substrates, thus greatly restricting their widespread applications. The nucleophilic addition of 1,3-enynes provides another way to allene moieties (Scheme 1a).[9] Nevertheless, the use of highly active and unstable organometallic reagents also hampers the substrate scope.

With the rapid growth of photo- and electrocatalytic synthesis in the past decade,[10] free radical chemistry has achieved remarkable progress, which thus has brought the radical addition of 1,3-enynes up to a new level. Peculiarly in combination with transition-metal catalysis, a variety of 1,4-difunctionalizations of 1,3-enynes have been successfully implemented (Scheme 1b).[11] However, thus far established radical precursors are primarily limited to alkyl,[12] perfluoroalkyl,[13] trifluoromethyl,[14] and sulfonyl[15] radical sources. Other types of radical addition have been rarely studied, but at all events, their development is a very appealing research aspect to lead the radical addition reaction of 1,3-enynes into synthetic maturity.

On the other hand, the introduction of silicon atoms can improve the lipophilicity of organic molecules, modify their physicochemical properties, particularly enhance the tissue permeability of drugs, and prolong their action time. Thus, silicon substitution plays an important role in the innovation

Scheme 1. Methodologies for the Synthesis of Allenes from 1,3-Enynes



and development of new drugs.[16] Moreover, organosilicons represent a class of irreplaceable synthons in numerous classical reactions.[17] Therefore, selective installation of complex organosilicon molecules has become a promising area.[18] In recent years, the rise of photoinduced silyl radical

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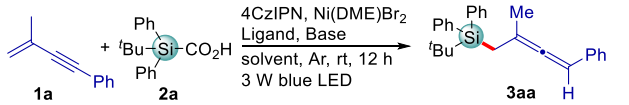
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chemistry has provided a sustainable and efficient protocol for C−Si bond formation. Initially, silyl radicals were generated through the cleavage of Si−X (X = Hg, Se, B, P, Si, etc.) bonds under ultraviolet (UV) irradiation.[19] However, such high-energy irradiation inevitably causes the formation of undesired byproducts. Direct activation of Si−H bonds under visible light irradiation to produce silyl radicals always requires additionally an expensive hydrogen atom transfer (HAT) catalyst.[20] Thus indisputably, silanecarboxylic acids are ideal silyl radical reagents that are easily available, highly stable, and easy to operate. Moreover, their chemical properties are analogous to those of carboxylic acids, where radical decarboxylation readily occurs under mild and concise conditions. Heretofore, decarboxylative fictionalizations of silanecarboxylic acids have been scarcely studied, and only several examples, such as alkylation,[21] allylation,[22] difluoroallylation,[23] and heterocyclic arylation,[24] were documented. Intrigued by this, we report a regioselective hydrosilylation of 1,3-enynes with silanecarboxylic acids as the silylating sources. This reactivity is especially synthetically useful when performed under MeOD as the solvent, where deuterosilylation products were obtained with high deuterium isotopic contents.

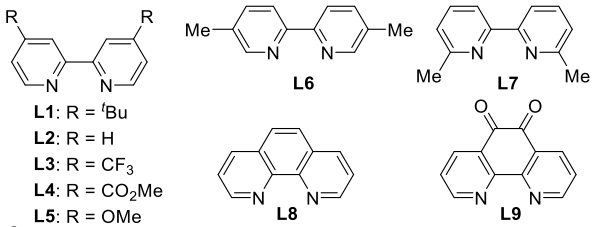
Our optimization was initiated by the use of (3-methylbut-3-en-1-yn-1-yl)benzene (**1a**) and *tert*-butyldiphenylsilanecarboxylic acid (**2a**) as the model reaction. Screening of catalysts showed that the dual catalytic system with $Ni(DME)Br_2$ and 2,4,5,6-tetra(9H-carbazol-9-yl)isophthalonitrile (4CzIPN) furnished the desired hydrosilylation product in 58% yield (Tables S1 and S2 of the Supporting Information). Different solvents were tested, among which originally used MeOH led to a higher yield than the others (entries 1−6 in Table 1). Switching the base from K_2HPO_4 to K_2CO_3, K_3PO_4, or KOAc gave inferior yields, indicating that K_2HPO_4 was the most powerful for this conversion (entries 7−9 in Table 1). A further ligand evaluation suggested that the efficiency of dtbpy (**L1**) was higher than that of other ligands (entries 10−17 in Table 1). Only an 18% yield was detected in the absence of a nickel catalyst (entry 18 in Table 1). To our delight, the yield could be improved to 83% by increasing the loading of compound **2a** (entry 19 in Table 1). A control experiment pointed out that photoirradiation is indispensable (entry 20 in Table 1).

Under the optimized reaction conditions, the scope of 1,3-enynes with compound **2a** as the coupling partner was evaluated (Table 2). The reactivity showed good functional group tolerance. For instance, both electron-donating groups, such as methyl, *n*-propyl, *tert*-butyl, and phenyl groups, and electron-withdrawing groups, such as methoxycarbonyl, trifluoromethyl, cyano groups, and halogen, were well-compatible (**3aa**−**3ma**). 4-Alkyl-substituted 1,3-enyne was suitable but resulted in an inferior yield (**3na**). Only a 26% yield was obtained from internal 1,3-enyne (**3oa**), implying that the reactivity of internal 1,3-enyne is lower than that of terminal substrates. 2-Aryl-substituted 1,3-enyne was also investigated. To our satisfaction, the target product was delivered in a moderate yield (**3pa**). Subsequently, several silanecarboxylic acids were explored to prove the widespread applicability of this transformation. It was found that methyldiphenylsilanecarboxylic acid proceeded smoothly and led to the desired product efficiently (**3ab**). However, methyldiphenylsilanecarboxylic acids with a methyl or *tert*-butyl group at the *meta* or *para* position of the benzene ring exhibited low reactivity (**3ac**−**3ae**). Note that methyldiphenylsilanecarboxylic acid bearing a methyl group at the *ortho* position of the benzene ring was inapplicable to this conversion, probably as a result of high steric hindrance. Dimethylphenylsilanecarboxylic acid and triphenylsilanecarboxylic acid also showed inefficiency under the current catalytic system.

Table 1. Optimization of the Reaction Conditions[a]



entry	solvent	base	ligand	**3aa** (%)[b]
1	MeOH	K_2HPO_4	**L1**	58
2	DMF	K_2HPO_4	**L1**	0
3	DMSO	K_2HPO_4	**L1**	0
4	MeCN	K_2HPO_4	**L1**	44
5	acetone	K_2HPO_4	**L1**	49
6	THF	K_2HPO_4	**L1**	23
7	MeOH	K_2CO_3	**L1**	41
8	MeOH	K_3PO_4	**L1**	53
9	MeOH	KOAc	**L1**	26
10	MeOH	K_2HPO_4	**L2**	24
11	MeOH	K_2HPO_4	**L3**	10
12	MeOH	K_2HPO_4	**L4**	4
13	MeOH	K_2HPO_4	**L5**	10
14	MeOH	K_2HPO_4	**L6**	21
15	MeOH	K_2HPO_4	**L7**	0
16	MeOH	K_2HPO_4	**L8**	14
17	MeOH	K_2HPO_4	**L9**	0
18[c]	MeOH	K_2HPO_4	**L1**	18
19[d]	MeOH	K_2HPO_4	**L1**	83 (75)
20[d,e]	MeOH	K_2HPO_4	**L1**	0



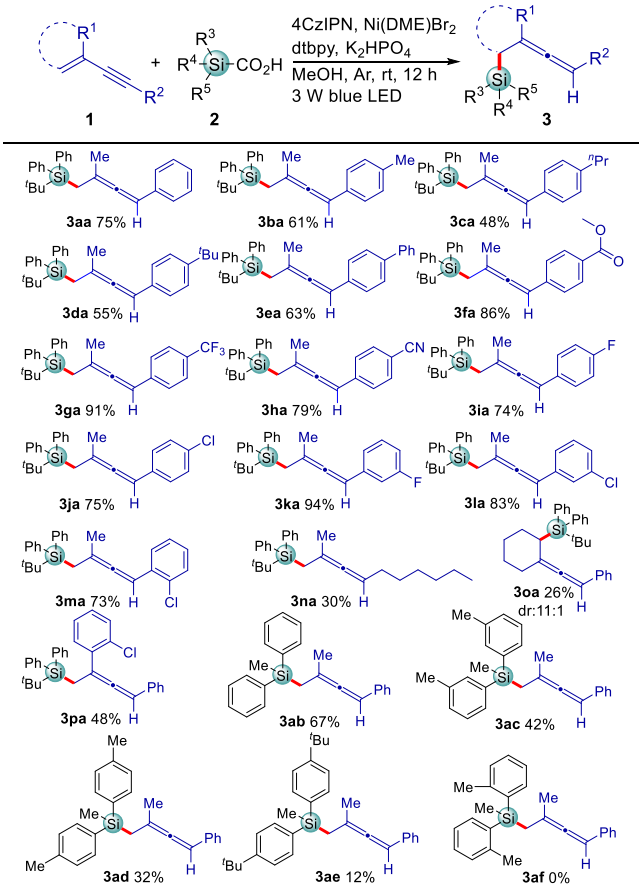
[a]Conditions: 0.2 mmol of compound **1a**, 1.0 equiv of compound **2a**, 2.5 mol % 4CzIPN, 10 mol % $Ni(DME)Br_2$, 10 mol % ligand, 0.5 equiv of base, 2 mL of solvent, and room temperature, under an argon atmosphere at a distance of 1 cm from a 3 W blue light-emitting diode (LED) for 12 h. [b]Yields determined by ^1H nuclear magnetic resonance (NMR) using 1,3,5-trimethoxybenzene as the internal standard, with the isolated yield in parentheses. [c]Without $Ni(DME)Br_2$. [d]A total of 1.8 equiv of compound **2a** was used. [e]In the dark. DME = 1,2-dimethoxyethane.

An infinitely better application of this transformation in organic synthesis is that it can be used for assembling α-silyldeuteroallenes. Amazingly, the deuterosilylation of 1,3-enynes could proceed smoothly just by switching the solvent to MeOD, leading to good yields as well as high deuterium isotopic contents (Table 3). Substrates converted similarly with hydrosilylation. However, the substrate bearing a methoxyl group as the strong electron-rich group gave only a 27% yield (**4qa**), which suggested that electron-deficient groups were in favor of this conversion.

This reaction can be successfully scaled up to 2 mmol for the example of compound **3aa** (Scheme 2). Several control experiments were carried out to shed light on the reaction

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Table 2. Substrate Scope of Hydrosilylation of 1,3-Enynes[a,b]

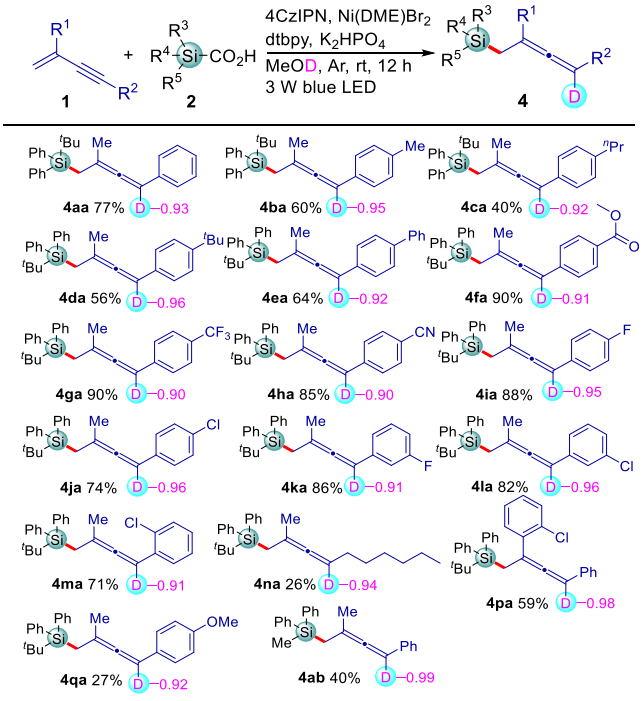


[a]Conditions: 0.2 mmol of compound **1**, 1.8 equiv of compound **2**, 2.5 mol % 4CzIPN, 10 mol % $Ni(DME)Br_2$, 10 mol % ligand, 0.5 equiv of K_2HPO_4, 2 mL of MeOH, and room temperature, under an argon atmosphere at a distance of 1 cm from a 3 W blue LED for 12 h. [b]Yields of isolated products were given.

pathway. A free-radical scavenger, such as (2,2,6,6-tetramethylpiperidin-1-yl)oxyl (TEMPO) or butylated hydroxytoluene (BHT), was added to the standard conditions. As expected, the desired conversion was outright inhibited (Scheme 3a), which meant that a radically associated process might be involved. The direct utilization of $MePh_2SiH$ as the silyl radical precursor was also attempted (reactions b and c of Scheme 3). However, no product was detected even with additionally a HAT catalyst or under UV irradiation, proving the unique reactivity of silanecarboxylic acids in the photo and nickel dual catalytic system.

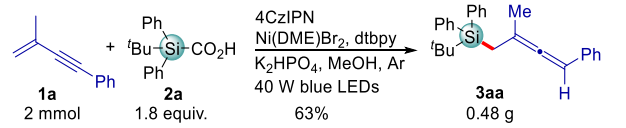
On the basis of previous reports and our above investigations, we proposed a possible mechanism, as elucidated in Scheme 4. A carboxyl radical **A** as the key species is formed by single electron transfer (SET) between silanecarboxylic acid and the excitation of 4CzIPN, simultaneously accompanied by the generation of the 4CzIPN radical anion. Then, decarboxylation of the carboxyl radical **A** released a new silyl radical intermediate **B**, which subsequently undergoes radical addition with 1,3-enyne to produce propargyl radical **C**. Propargylic radical **C** can isomerize to another resonance form, allenyl radical **D**, which readily intercepts the nickel catalyst to create allenyl nickel complex **E**.

Table 3. Substrate Scope of Deuterosilylation of 1,3-Enynes[a,b]

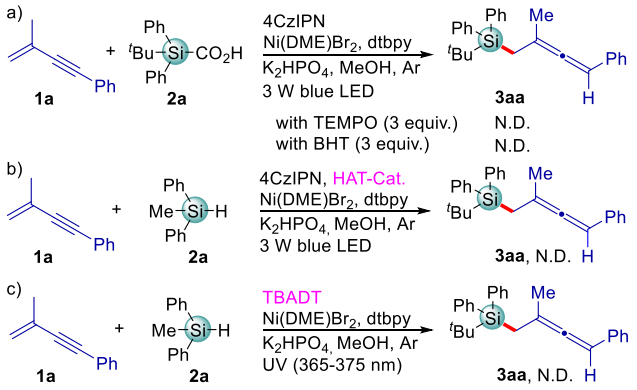


[a]Conditions: 0.2 mmol of compound **1**, 1.8 equiv of compound **2**, 2.5 mol % 4CzIPN, 10 mol % $Ni(DME)Br_2$, 10 mol % ligand, 0.5 equiv of K_2HPO_4, 2 mL of MeOD, and room temperature, under an argon atmosphere at a distance of 1 cm from a 3 W blue LED for 12 h. [b]Yields of isolated products were given.

Scheme 2. Gram-Scale Reaction



Scheme 3. Control Experiments

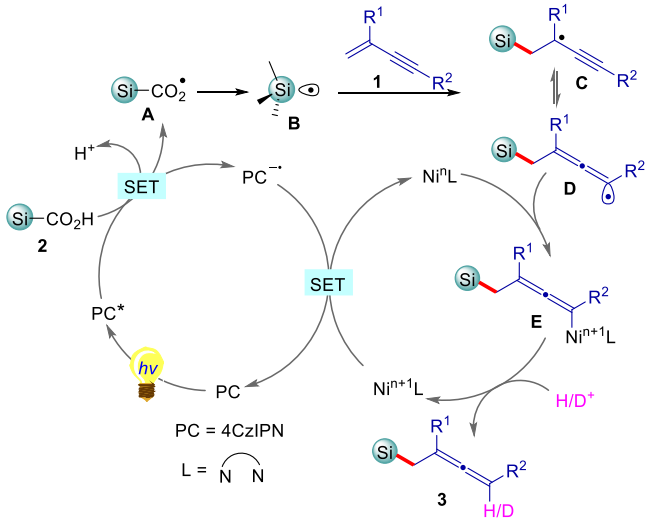


Finally, a protonation and another SET result in the formation of the eventual product **3**, along with the regeneration of the nickel catalyst and the ground-state 4CzIPN.

In summary, we have disclosed a photoredox and nickel cooperatively catalyzed hydrosilylation of 1,3-enynes, providing an efficient, concise, and modular strategy to access diverse substituted α-silylallene derivatives with high selectivity. In this

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Scheme 4. Possible Mechanism



transformation, silanecarboxylic acids as the silyl radical precursors not only have the advantages of high stability and easy operation but also exhibit unique reactivity in our photo/nickel dual catalytic system. The reaction variant is particularly synthetically valuable for the synthesis of α-silyldeuteroallenes when performed under MeOD as the solvent. Further investigations on the selective silicofunctionalization of 1,3-enynes are ongoing in our laboratory.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.3c04064.

Experimental procedures, characterization data for compounds, and NMR spectra of products (PDF)

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Notes

The authors declare no competing financial interest.

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